Computer Network Lab Questions (BCA)

1. IP Address Configuration

- Configure a static IP address on a Windows
- Verify the configuration using ipconfig (Windows)

2. Network Connectivity Testing

- Use the ping command to test connectivity between two computers.
- Record and analyze the results.

3. Traceroute Experiment

- Use tracert (Windows) to trace the path to a website.
- Identify the number of hops.

4. Peer to Peer connection

5. Basic Network Setup in cisco packet tracer

Objective: Connect two PCs using a switch.

Tasks:

1. Place 2 PCs and 1 Switch.
2. Connect devices using copper straight-through cables.
3. Assign IP addresses:
    - PC0: 192.168.1.10/24
    - PC1: 192.168.1.20/24
4. Test connectivity using ping.

6. Network Cable Creation

- Prepare a straight-through Ethernet cable using RJ-45 connectors.
- Test the cable using a cable tester.

7. **Switch Configuration in cisco packet tracer**

- Configure a switch with:

- Hostname
- Management IP address
- Password protection

8. **VLAN Configuration in cisco packet tracer**

- Create two VLANs on a switch.
- Assign ports to different VLANs.
- Test communication between VLANs.

9. Router Configuration

- Configure basic settings on a router:
  - Hostname
  - Password
  - Interface IP address
  - Save configuration

10. Inter Vlan connection using router in cisco packet tracer
Objective: Enable communication between VLANs.

Tasks:

1. Create VLANs.
2. Configure trunk port.
3. Configure Router-on-a-Stick.
4. Test communication.

11. Configure Router in Stick Model in cisco packet tracer

- Pass and test connection for 4 vlan through single port of Router

12. DHCP service Configuration in cisco packet tracer

- Configure a DHCP server.
- Verify automatic IP assignment to clients.

13. DNS Testing

- Use nslookup to find the IP address of a domain.
- Explain DNS resolution.

14. Configure network device using console cable

15. Configure SSH Connection

Objective: Secure remote access to a router.

Tasks:

1. Configure hostname and domain name.
2. Generate RSA keys.
3. Create local user.
4. Enable SSH.

16 Static Routing Configuration in cisco packet

Objective: Connect three networks using two routers.

Tasks:

1. Create three LANs.
2. Configure router interfaces.
3. Configure static routes.
4. Verify end-to-end connectivity.

17 . RIP Routing Configuration in cisco packet tracer

Objective: Configure RIP v2.

Tasks:

1. Connect 3 routers.
2. Assign IP addresses.

3. Configure RIP v2.

18. OSPF Routing Configuration in cisco packet tracer

19. NAT Configuration

- Configure Network Address Translation on a router.
- Verify internet access from private IP addresses.

20. Firewall Configuration

- Configure firewall rules to block and allow specific services.
- Test the results.

21. Network Troubleshooting

Using commands such as:

- ping
- ipconfig
- netstat
- nslookup
- tracert

Diagnose and solve a given network connectivity problem.

22. Design the topology of Namuna College Lab and configure.
23. Design the topology of Namuna College.